Doctoral Dissertation

Dynamic Machine Learning Modeling with ACI Shear data

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MISSOURI OF SCIENCE OF TECHNOLOGY

Doctorate of Engineering

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Doctoral Dissertation

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Acknowledgements

I would like to thank [...], for [...]

ABSTRACT

In many instances, data are kept in data science. We remove properties (columns) of instances, but rarely delete an instance itself, unless we do an outlier analysis. When dealing with imprecise data, such as shear test data from the American Concrete Institute (ACI), too many points can skew what would be the correct results. Having a dynamic, supervised, machine learning approach, which updates models when training and seeing new data, would both minimize storage or memory and behave robustly against outlier results. This approach combines multiple fields of ontology to create an empirical-experimental analysis.

Keywords: Dynamic, Machine Learning, Supervised Learning, ACI, Shear test data

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(Optional)

The List of Tables has a similar format to the table of contents. In case there are many tables within the text, it is recommended to automatically create a list of tables through Word as shown in the following example.

LIST OF TABLES

3.1 Enrollment in local colleges, 2005 3

(Optional) The List of Figures has a similar format to the table of contents. Figures, diagrams and other graphics appearing in the text are all referred as figures and follow uniform numbering. In case there are many figures within the work, it is recommended to automatically create a list of figures through Word as shown in the following example.

LIST OF FIGURES

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LIST OF ABBREVIATIONS

Important abbreviations that have been used in the text and need explanation are briefly presented.

ACI American Concrete Institute

1 Introduction

The introduction is the first section of the document and covers the 30% of the whole dissertation. It draws your reader in, setting the stage for your research with a clear focus, purpose, and direction on a relevant topic. It may include a review of the background theory in your subject area in the form of a literature review. The literature review may be in a separate section. [1] [**?**]. [**?**, **?**, **?**, **?**, **?**, **?**, **?**]

The introduction should include:

• Dissertation topic, in context: give necessary background information

• Your focus and scope: focus and define the scope of your research

• The relevance of your research: define how your research relates to existing work on your topic. Explain how it solves a practical or theoretical problem, addresses a gap in the literature, builds on existing research and proposes a new understanding of the topic.

• Your questions and objectives: explain what your research aims to find out, and how.

• An overview of your structure: end the introduction with an outline of the structure of your dissertation to follow. Share a brief summary of each chapter and show clearly how each contributes to your central aims.

1.1 Aims and Objectives

1.2 Research Questions

1.3 Contribution

1.4 Structure of the Thesis

1.5 Summary

2 Research Methodology

This section describes the materials and methods used for the research. It is essential to describe the steps clearly and accurately to enable readers to determine how much your methodology may have affected your results and to replicate your study if necessary. This allows them to evaluate your research's reliability and validity and dissertation topic.

A methods section may include an overview of the research design, details of the sample and the sampling technique used, and a description of procedures, materials, and statistical tools.

2.1 Section

2.2 Section

2.3 Summary

3 Results/ Findings

The results section presents the main findings of your research. Report all relevant results concisely and objectively, in a logical order. That gives your reader a clear idea of exactly what you found and keeps the data itself separate from your subjective analysis. Any evaluation of the findings should be discussed in the next section.

This section may contain visual elements accompanied by text. In quantitative research, it's often helpful to include graphs, charts, tables, etc., but only if they are directly relevant to your results. Give these elements clear, descriptive titles and labels so that your reader can easily understand what is being shown (use numbering and caption title as recommended).

Tables must be as simple as possible. Use numbers and title headings for each one. See the example below:

Table 3.1: Enrollment in local colleges, 2005

College	New students	Graduating students	Change
	Undergraduate		
Cedar University	110	103	+7
Elm College	223	214	+9
Maple Academy	197	120	+77
	Graduate		
Cedar University	24	20	+4
Elm College	43	53	-10
Maple Academy	3	11	-8
Total	600	521	79

Use the word "Figure" for images, diagrams and charts. Use numbers and title headings for each one. See the example below:

3.1 Section

3.2 Section

3.3 Summary

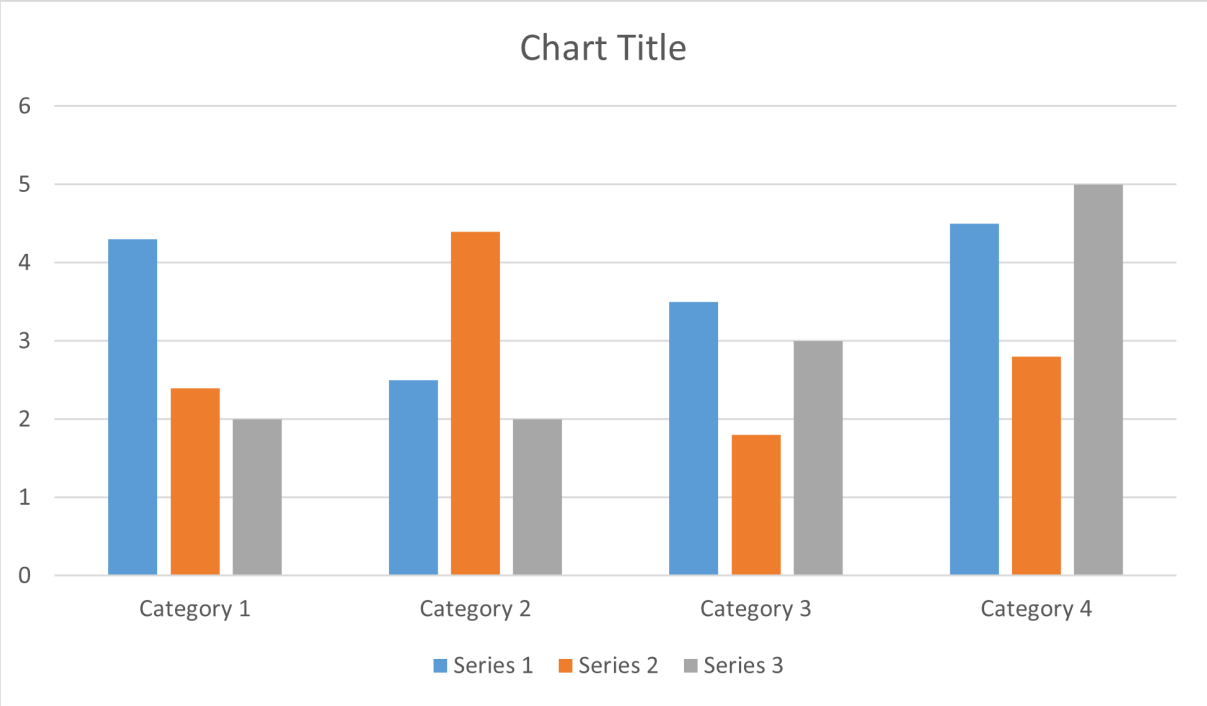

Figure 3.1: Enrollment in local colleges, 2005

4 Discussion/ Interpretation

In this section, you discuss your results, while writing a clear statement as to whether or not the results support the original hypotheses or research question. Compare your findings with those of other research dealing with the same or similar topic. Then, give the differences and similarities between them and specify what new evidence or knowledge emerges from this research.

4.1 Section

4.2 Section

4.3 Summary

5 Summary of Findings/ Recommendations

In this section, you remind the purpose of the study, provide a summary of the results and report the conclusions derived from the study.

Here, you should mention the limitations of the study, related to the formulation of research aim and objectives, application of data collection method, sample size, the scope of discussions and analysis, etc. You can propose future research suggestions that address the limitation of the study and the generalizability of the results.

Finally, include suggestions for further research. Additionally, you can propose future studies that can address the same research problem in a different setting, context, location and/or culture.

5.1 Section

5.2 Summary

BIBLIOGRAPHY

[1] R. Piraei, M. Niazkar, S. H. Afzali, and A. Menapace, “Application of machine learning models to bridge afflux estimation,” *Water*, vol. 15, no. 12, p. 2187, 2023. [Online]. Available: https://doi.org/10.3390/w15122187

APPENDICES

APPENDIX I

Title of Appendix

If you have material that cannot be included within your document, you must include an appendix. You may include one appendix or a number of appendices. If you have more than one appendix, you would number each accordingly (i.e., Appendix I, Appendix II, etc.). Write your appendix headings in the same manner as your chapter headings.